# Protons, Neutrons & Electrons

**Exam code:** 0625 & 0972

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## Relative charge

- **Charge** can be positive or negative
- Protons, neutrons and electrons have different properties and different charges
  - **Protons** have a **positive** charge
  - **Electrons** have a **negative** charge
  - **Neutrons** have **no charge**
- **Relative charge** can be used to compare these properties
  - Relative charge is the **ratio** of the charge of a particle compared to the fundamental charge
  - Because relative charge is a ratio, it has **no units**
- **Fundamental charge** is equal to the **magnitude** of the charge on a proton and an electron = **$1.6 \times 10^{-19}$ C**
- The relative charge of a proton is

$$\text{proton} = \frac{1.6 \times 10^{-19}\ \text{C}}{1.6 \times 10^{-19}\ \text{C}} = +1$$

- The relative charge of an electron is

$$\text{electron} = \frac{-1.6 \times 10^{-19}\ \text{C}}{1.6 \times 10^{-19}\ \text{C}} = -1$$

- The relative charge of a neutron is

$$\text{neutron} = \frac{0\ \text{C}}{1.6 \times 10^{-19}\ \text{C}} = 0$$

### Table of relative charges

| Particle | Relative charge |
|---|---|
| Proton | +1 |
| Neutron | 0 |
| Electron | –1 |

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

### Examiner Tips and Tricks

particle.

# Nuclear charge

**Extended tier only**

- Nuclear charge is the **relative charge** of the **nucleus**
- This is determined by the **proton number** of the atom
- Relative charge is calculated by:

$$\text{nuclear charge} = \text{number of protons in the nucleus} \times \text{relative charge}$$

- An element of carbon has a proton number of 6
  - So the relative charge of carbon is 6 × (+1) = 6

## Worked Example

What is the relative charge of the chromium nucleus $^{52}_{24}\text{Cr}$?

**Answer:**

**Step 1: Determine the number of protons**

- The number of protons is the proton number
- This is the Z number in the AZX notation

$$^{A}_{Z}\text{X}$$

- This chromium nucleus has 24 protons

$$^{52}_{24}\text{Cr}$$

**Step 2: State the relative mass of 1 proton**

- 1 proton has a relative charge of +1

**Step 3: Multiply relative charge of 1 proton by the number of protons**

- This nucleus of chromium therefore has a relative charge of +24

## Examiner Tips and Tricks

Charge can be either positive (+) or negative (−). Usually, when a number is positive, we don't write the + sign, but when dealing with relative charge, we do.

**Extended tier only**

- Nuclear mass or the relative mass of an atom, is the **relative mass** of the **nucleus**
  - This is because the mass of electrons orbiting the nucleus is negligible
- A relative mass of 1 = $1.67 \times 10^{-27}$ kg
- The relative mass of an atom is determined by the **nucleon number** of the atom
- It is calculated by:

$$\text{relative mass of atom} = \text{number of protons and neutrons in the nucle}$$

- An element of carbon has a nucleon number of 12
  - So the relative mass of carbon is 12 × 1 = 12 atomic mass units or **amu**

## Table of relative mass

| Particle | Relative mass |
|---|---|
| Proton | 1 |
| Neutron | 1 |
| Electron | $\frac{1}{2000}$ negligible |

### Worked Example

What is the relative mass of the chromium nucleus $^{52}_{24}\text{Cr}$?

**Answer:**

**Step 1: Determine the number of protons and neutrons**

- The number of protons and neutrons is the mass (nucleon) number
- This is the A number in the AZX notation

$$^{A}_{Z}\text{X}$$

- This chromium nucleus has 52 protons and neutrons

$$^{52}_{24}\text{Cr}$$

**Step 2: State the relative mass of 1 proton and 1 neutron**

- 1 proton has a relative mass of 1
- 1 neutron has a relative mass of 1

**Step 3: Multiply relative charge of 1 proton and 1 neutron by number of protons and neutrons**

### Examiner Tips and Tricks

The relative mass of a nucleus only includes the protons and neutrons. However, this is pretty much the relative mass of the whole atom because electrons have negligible (very little) mass in comparison to the proton and neutron.

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